Code No: 156AM **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February/March – 2022
CONCURRENT PROGRAMMING
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Describe the properties of program correctness.
b) Illustrate the concept of shared objects and synchronization between concurrent threads for prime number generation modules. [5+10]

2.a) What is parallelism? State the need for parallelism.
b) Explain the Filter-Lock algorithm for implementing mutual exclusion among n-threads. [5+10]

3.a) Recall a simple lock-based concurrent FIFO queue. Write enq() and deq() methods and examine how these methods results in correctness property with a brief illustrations.
b) What are sequential objects? Discuss briefly. [10+5]

4.a) Explain how sequential consistency is non-blocking with relevant example.
b) Give a detail note on a dependent non-blocking progress condition. [8+7]

5.a) "Any nontrivial RMW register has consensus number at least 2". Prove the theorem.
b) Explain the following operations with brief illustrations: [7+8]
i) compareAndSet() ii) getAndSet()

6. Write down the lock-free universal algorithm and explain the execution of the lock-free universal construction with a brief illustration. [15]

7.a) State the merits and demerits of hand-over-hand locking mechanism.
b) How to implement Optimistic synchronization mechanism on a linked list by concurrent threads? Write the complete pseudo code for add and remove methods of it and explain. [5+10]

8. What do you mean by lock-free stack? Explain how to implement lock-free stack operations with an example program code. [15]

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